

AI Engineer Roadmap

Relevant for roles like: AI Engineer, Gen AI Engineer, LLM Engineer, Agentic AI Engineer, AI Agent Engineer, Forward Deployed Engineer, AI Solutions Architect, AI Platform Engineer, Applied AI Engineer, MLOps Engineer, LLMOps Engineer.

Follow the steps in order — each one builds on the last. Don't skip Steps 1-2 even if you're in a hurry.

Step 0: Must Know

Six terms that show up in every AI engineering conversation: LLM, RAG, MCP, Agent, Fine-tuning, Quantization.

Step 1: Machine Learning Foundations

What ML is, how machines learn, and the vocabulary used everywhere else.

- What is Machine Learning?
- Supervised vs Unsupervised Learning
- Linear Regression vs Logistic Regression
- Feature Engineering
- Precision vs Recall
- L1 and L2 Loss Functions
- Regularization (L1 vs L2)
- Reinforcement Learning
- Contrastive Learning

Step 2: Deep Learning and Neural Networks

How a neural network actually learns — gradient descent, backprop, and stable-training tricks.

- Bias in Artificial Neural Networks
- Gradient Descent (the math)
- Backpropagation (the math)
- Cross-Entropy Loss
- Dropout
- Batch Normalization vs Layer Normalization
- RMSNorm
- Recurrent Neural Networks (RNN)
- How PyTorch Works
- How TensorFlow Works

Step 3: Generative AI and the Transformer Architecture

Generative AI and the Transformer from the inside — tokens, embeddings, attention.

- What is Generative AI?
- What happens inside ChatGPT after you hit Enter
- Autoregressive Models
- Byte Pair Encoding (BPE)
- Tokenization
- Embeddings
- RNNs vs Transformers
- Transformer Architecture
- Encoder vs Decoder
- Self Attention
- Attention math (Q, K, V)
- Why attention is scaled by $\sqrt{d_k}$
- Causal Masking
- Multi-Head Attention
- Cross Attention
- Positional Embeddings
- RoPE (Rotary Position Embedding)
- Feed-Forward Network in LLMs

Step 4: How LLMs Generate Text

Next-token selection, creativity controls, streaming, and where context windows break down.

- Temperature
- Top-k and Top-p Sampling
- Token Streaming
- Why Context Windows Are Limited
- Lost in the Middle Problem

Step 5: Modern LLM Architecture

What modern LLMs add on top of the basic Transformer to scale further and go faster.

- Mixture of Experts (MoE)
- Grouped Query Attention (GQA)
- Sliding Window Attention
- Attention Sinks
- Flash Attention

- DeepSeek-V4 Architecture (case study)

Step 6: Types of Language Models

Beyond the standard large text-generating LLM.

- Small Language Models (SLMs)
- Large Reasoning Models (LRMs)
- Recursive Language Models (RLMs)
- Diffusion Language Models (DLMs)
- Jev and System One Models

Step 7: Training, Fine-Tuning, and Alignment

Adapting a pre-trained model to a task, shrinking it, and teaching it to follow instructions.

- Fine-Tuning
- LoRA (Low-Rank Adaptation)
- Prefix Tuning
- Knowledge Distillation
- Continual Learning & Catastrophic Forgetting
- Deep RL from Human Preferences (the paper behind RLHF)
- InstructGPT
- RLHF
- Proximal Policy Optimization (PPO)
- Direct Preference Optimization (DPO)
- Group Relative Policy Optimization (GRPO)

Step 8: Prompt Engineering and Context Engineering

Getting better answers out of an LLM, and managing what goes into its context window.

- Chain-of-Thought (CoT) Prompting
- Prompt Chaining
- Prompt Caching
- Context Engineering
- Context Compaction

Step 9: Vector Search and RAG

Giving an LLM knowledge it wasn't trained on — from vector storage to advanced RAG variants.

- Vector Databases
- Approximate Nearest Neighbor (ANN) Search

- Semantic Search
- Hybrid Search
- Rerankers
- ColBERT (late interaction retrieval)
- Chunking Strategies
- HyDE
- Embedding Cache
- Semantic Caching
- Agentic RAG
- GraphRAG
- Vectorless RAG

Step 10: AI Agents and Agentic Systems

From answering questions to doing work — tools, memory, and multi-agent systems.

- What is an AI Agent?
- Function Calling
- AI Agent Loop
- ReAct Agent
- Plan-and-Execute Agent
- Reflection Agent
- AI Agent Memory
- MCP (Model Context Protocol)
- Agent Skills
- OKF (Open Knowledge Format)
- Multi-Agent Systems
- AI SubAgents
- How AI Agents Communicate
- AI Orchestration
- Sakana Fugu (technical case study)
- Computer-Use Agents

Step 11: Agentic Engineering and Agent Frameworks

Engineering practices for reliable agents, plus how popular frameworks and coding agents work.

- Harness Engineering
- Loop Engineering
- Graph Engineering

- Defining "done" for AI systems
- LangChain
- LangGraph
- Claude Code (how it works)
- Cursor (how it works)

Step 12: LLM Inference Engineering

Making LLMs faster and cheaper to run: caching, batching, speculation, quantization, serving engines.

- LLM Inference Optimization
- Prefill vs Decode
- First-Token Latency Problem
- Prefill-Decode Disaggregation
- KV Cache
- KV Cache Compression
- Paged Attention
- Continuous Batching
- Speculative Decoding
- N-gram Speculation
- Medusa (multi-head speculative decoding)
- EAGLE (feature-level speculative decoding)
- Model Quantization
- GGUF
- llama.cpp
- vLLM
- SGLang
- TensorRT-LLM

Step 13: Evaluation and Observability

Measuring whether your LLM and agent are actually doing a good job, in production.

- LLM Evaluation: Metrics, Benchmarks, Methods
- LLM as a Judge
- Evaluating AI Agents
- AI Agent Observability (traces, spans, metrics)

Step 14: AI Safety and Security

Keeping an LLM app safe, how attackers try to break it, and detecting AI-generated text.

- LLM Guardrails
- Prompt Injection: Attacks and Defenses
- LLM Watermarking

Step 15: Multimodal AI and Generative Models

Working with images and other data types, plus the generative models behind image synthesis.

- Multimodal AI
- Vision Transformer (ViT)
- Image Embeddings
- Diffusion Models
- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)

Step 16: AI Infrastructure, Deployment, and System Design

Hardware, deployment targets, routing, and end-to-end AI system design.

- How a GPU Works for Deep Learning
- How a Google TPU Works
- How an LPU Works
- Cloud vs On-device Model Deployment
- On-device ML (e.g. TensorFlow Lite)
- LLM Routing
- Designing a Real-Time Voice AI Agent
- (Supporting) System Design fundamentals
- (Supporting) HTTP Request vs Long-Polling vs WebSocket vs SSE
- (Supporting) How Voice and Video Calls Work

Step 17: Frontier Ideas in AI

Where the field is heading — world models, self-improving systems.

- Joint Embedding Predictive Architecture (JEPA)
- World Models
- Recursive Self-Improvement (RSI)

Step 18: Prepare for AI Engineering Interviews

- AI Engineering interview questions and answers (system design, ML fundamentals, LLMs, agents)